Joshua Kim

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EDUCATION

University of California, Santa Barbara (UCSB), Santa Barbara, CA **June 2024**

Master of Science (M.S.), Computer Engineering

Relevant Coursework: AI/ML in Test and Design Automation, Pattern Recognition, Distributed System, Natural Language Processing, Mobile Embedded System, Advanced Computer Architecture, Trustworthy ML in Security, Quantum Photonics

University of California, Santa Barbara (UCSB), Santa Barbara, CA **June 2023**

Bachelor of Science (B.S.), Computer Engineering

Relevant Coursework: Object-Oriented Design, Intro to Computer Architecture, Data Structure & Algorithm, Intro to Computer Vision, Advanced App Program, Sensor and Peripheral Interface Design, Computer Security

TECHNICAL SKILLS

- Python, PyTorch, C/C++, Java, Matlab, Mathematica, Git, HTML, CSS, JavaScript, CI/CD, XML, Quantum Photonic, CUDA, AWS, GCP, Docker, x86, MIPS,STM32, Ghidra, Linux, SQL, Rust, NVMe

TEACHING/WORK EXPERIENCE

GPU Software Engineer | Graid Technology Inc, Sunnyvale, CA **October 2025 - Present**

- Design a Key Value Store for cuda library that will be used for cuda application and storage
- Integrate KV cache with the existing product to speed up AI inference
- Benchmark and profile using cuda-gdb, ncu, nsys, cuobjdump for cuda applications
- Researching, Designing new features through software development life cycle

***Embedded System Teaching Assistant* | UCSB, Goleta, CA** **January 2024-June 2024**

- Design microcontroller based circuits within an integrated development environment that interfaces with external peripherals.
- Instructed serial and peripheral interfaces including GPIO, UART, I2C, SPI, USB and communication interfaces including Wi-Fi, Bluetooth, and GPS.
- Educated how ADC and DAC interface with external circuits, as well as digital signal processing concepts such as sampling and digital filters.
- Directed non-blocking motor and UART communication on an STM32L4 Nucleo Board, ensuring efficient operation without CPU interruptions. Integrated stepper motor, ADXL345 accelerometer, and TC-74 temperature sensor for precise door control.

***Computer Vision Teaching Assistant* | UCSB, Goleta, CA** **September 2023-December 2023**

- Designed challenging and instructive homework assignments that aligned with the curriculum's learning objectives in computer vision.
- Enhanced the students' understanding of computer vision principles and algorithms.
- Utilized strong communication and interpersonal skills to facilitate an effective learning environment during lectures and office hours.
- Provided guidance on leveraging PyTorch for deep learning to implement computer vision models.

***Software Engineer Intern* | MOOG, Goleta, CA** **June 2022-September 2022**

- Debugged, power cycled, and applied the knowledge of Embedded System and C programming from software to hardware.
- Coded in C for system functionality and Java for the GUI, enabling customers to interact easily with the interface.
- Engaged in group meetings by sharing ideas and experienced agile retrospective and code review that brought improvement.
- Developed expertise in using TortoiseSVN for version control, focusing on commit and checkout operations.

PROFESSIONAL PROJECT EXPERIENCE

***Modeling CACTI* | Computer Architecture, UCSB, Goleta, CA** **November 2023-December 2023**

- Developed machine learning models to simulate CACTI’s functionality, addressing its limitations in speed and technology node predictions.
- Implemented and evaluated 4 different models: KNN, MLP, SVM, and random forest.
- Achieved significant speed improvements with MLP processing over 1000 queries in 0.00043 seconds.
- Automated the data collection process, generating 9324 data points by varying cache configurations.

***Ratedar* | Mobile Embedded System, UCSB, Goleta, CA** **November 2023-December 2023**

- Developed an app allowing users to select, rate, and review various items, with the ability to share and view these items' locations.
- Implemented GPS-based relative distance functionality, enhancing user engagement by allowing them to see their proximity to rated items.
- Integrated Firebase for user authentication and real-time updates, enabling dynamic content management and secure user interactions.
- Designed a user-friendly interface and incorporated camera functionality, allowing users to add images alongside their reviews and comments.

Transfer Learning on PINN **| Tensor Computation UCSB, Goleta, CA June 2023-December 2023**

- Utilized tensor compression techniques to reduce the memory footprint of a pre-trained partial differential equation model and subsequently fine-tuned it for another partial differential equation, optimizing model efficiency and accuracy.
- Applied tensor computation techniques to reduce memory space and enhance faster computation, resulting in improved efficiency in data processing and model inference.
- Debugged neural network architectures, identifying and resolving issues in model training and validation, resulting in enhanced model performance and reliability.

Defect Detect **|CE Capstone Project, UCSB, Goleta, CA September 2022-June 2023**

- Designed an anomaly detection model that separates the almond from defective objects.
- Automate the process of synthesizing 30000 Almond pictures for training and validating the dataset.
- Implemented embedded C programming with an encoder to keep track of the position when objects pass through conveyor belt.

PROFESSIONAL CERTIFICATION

- CompTIA Security+ **March 2025**